

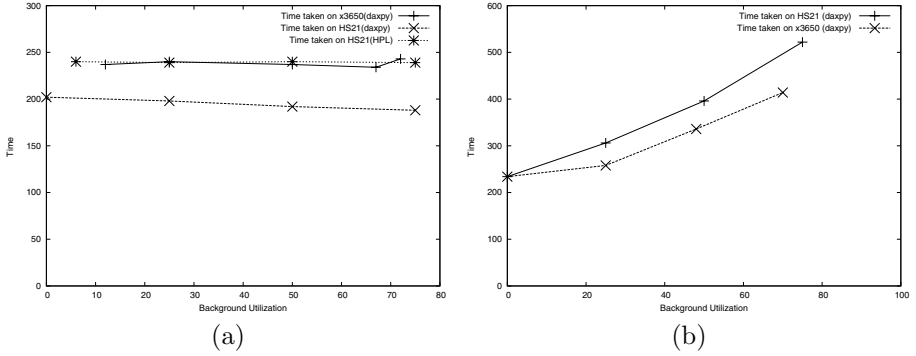


Fig. 2. Running Time of 30 Billion *daxpy* operations with change in background load on HS-21 and x3650 systems at (a) low memory footprint and (b) high memory footprint

both the foreground *daxpy* and background *fma*. We observed that as the size of the arrays being operated exceeded the L2 cache size (4MB), the applications were no longer isolated (Fig. 2(b)). The throughput of the foreground application decreases with increase in background traffic as a result of the large number of cache misses, which are due to increased cache usage by the background application. However, as one increases the memory footprint of each application beyond the cache size, the applications are no longer able to use the cache even in isolation. Hence, we concluded that for a large range of application use (small and large working set sizes), virtualization is able to successfully isolate two VMs from each other.

3.2 Migration Cost Modeling

We have proposed the application placement problem as a continual optimization problem, where we dynamically migrate the live virtual machines from one physical server to another in order to optimize the allocation. The migration of virtual machines requires creation of a checkpoint on secondary storage and retrieval of the VM image on the target server. Applications can continue to run during the course of migration. However, the performance of applications is impacted in the transition because of cache misses (hardware caches are not migrated) and possible application quiesces. Thus, each migration is characterized by a migration duration and a migration cost. The *Migration Manager* in *pMapper* needs to estimate this migration cost for use by the *Arbitrator*. Hence, we next studied the feasibility of characterizing migration cost for an application and study the parameters that affect this cost.

We observed (Fig. 3) that the impact of migration was independent of the background load and depends only on the VM characteristics. Hence, the cost of each live migration can be computed *a priori*. This cost is estimated by quantifying the decrease in throughput because of live migration and estimating the revenue loss because of the decreased performance (as given by SLA). Hence,